Kernel: R (system-wide)

```
#identify directory
my_wd <-getwd()
my_wd
```

Out[3]:
```
[1] "/home/user/Capstone-Project-Template"
```

In [4]:
```
#view available files
my_files <- dir(my_wd)
my_files
```

Out[4]:
```
 [1] "2026-04-05-folder-1"
 [2] "2026-04-05-notebook-1.ipynb"
 [3] "2026-04-05-notebook-2.ipynb"
 [4] "AdaptiveCapacity_natBreakNorm.lyr"
 [5] "AdaptiveCapacity_standardNorm.lyr"
 [6] "adaptiveCapacity.cpg"
 [7] "adaptiveCapacity.dbf"
 [8] "adaptiveCapacity.prj"
 [9] "adaptiveCapacity.sbn"
[10] "adaptiveCapacity.sbx"
[11] "adaptiveCapacity.shp"
[12] "adaptiveCapacity.shx"
[13] "Capstone-Project-Template.ipynb"
[14] "Connecticut_CAMA_and_Parcel_Layer_-6207911227706244147.csv.partialupload-
g4yGUAtfah"
[15] "geomen_cviBG.shp"
[16] "metadata_adaptiveCapacity.docx"
[17] "rosm.cache"
[18] "Stamford addresses geotagged.csv"
[19] "Stamford_CAMA_2025.csv"
[20] "zipped_layers_CVI.zip"
```

In [5]:
```
#load grand list data
data <- read.csv("Stamford_CAMA_2025.csv")
```

In [6]:
```
#confirm grand list loaded
head(data)
```

Out[6]:
```
  Parcel.ID Account.Number GIS.Tag     Map     Map.Cut Block Block.Cut Lot
1 16276     003-8541       W 002 3042      003 NA      8541  NA
2 16277     003-8542       W 003 3042      003 NA      8542  NA
3 16278     003-8543       W 004 3042      003 NA      8543  NA
4 16279     003-8544       W 005Z 3042     003 NA      8544  NA
5 16280     003-8545       E 006Z 3042     003 NA      8545  NA
6 16281     003-8546       E 001 3044      003 NA      8546  NA
  Lot.Cut Unit    ⋯ Sale.Grantor.Name        Instrument Validity.Code
1                 ⋯ KUHN ELIZABETH PATRICIA E 00         I
2                 ⋯ KURZ ELLIOT L ET AL       00         I
3                 ⋯ BANDER STEVEN ET AL       00         I
4                 ⋯ STEVENSON THOMAS P        00         I
5                 ⋯ OMALLEY MARY ALICE        00         I
```

```
6                  ⋯ SAZAN GUITA A             25         I
  Occupancy Condo.Main.ID CAMA.Site.Link
1 1                       https://gis.vgsi.com/stamfordct/Parcel.aspx?Pid=16276
2 1                       https://gis.vgsi.com/stamfordct/Parcel.aspx?Pid=16277
3 1                       https://gis.vgsi.com/stamfordct/Parcel.aspx?Pid=16278
4 1                       https://gis.vgsi.com/stamfordct/Parcel.aspx?Pid=16279
5 1                       https://gis.vgsi.com/stamfordct/Parcel.aspx?Pid=16280
6 1                       https://gis.vgsi.com/stamfordct/Parcel.aspx?Pid=16281
  Building.Photo
1 http://images.vgsi.com/photos/stamfordctPhotos//\\0153\\DSC02680_153632.JPG
2 http://images.vgsi.com/photos/stamfordctPhotos//\\0153\\DSC02678_153634.JPG
3 http://images.vgsi.com/photos/stamfordctPhotos//\\0153\\DSC02677_153636.JPG
4 http://images.vgsi.com/photos/stamfordctPhotos//\\0153\\DSC02676_153637.JPG
5 http://images.vgsi.com/photos/stamfordctPhotos//\\0153\\DSC02675_153640.JPG
6 http://images.vgsi.com/photos/stamfordctPhotos//\\0153\\DSC02738_153689.JPG
  Number.of.Units
1 NA
2 NA
3 NA
4 NA
5 NA
6 NA
  Sketch
1 https://gis.vgsi.com/stamfordct/ParcelSketch.ashx?pid=16276&bid=16276
2 https://gis.vgsi.com/stamfordct/ParcelSketch.ashx?pid=16277&bid=16277
3 https://gis.vgsi.com/stamfordct/ParcelSketch.ashx?pid=16278&bid=16278
4 https://gis.vgsi.com/stamfordct/ParcelSketch.ashx?pid=16279&bid=16279
5 https://gis.vgsi.com/stamfordct/ParcelSketch.ashx?pid=16280&bid=16280
6 https://gis.vgsi.com/stamfordct/ParcelSketch.ashx?pid=16281&bid=16281
  Land.Use.Note
1
2
3
4
5
6
```

In [7]:

```
#load DPLYR
library(dplyr)
```

Out[7]:

```
Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union
```

In [8]:

```
#Set CPACE compatible use codes
use_codes <- list("Com Condo Option", "Comm Condo  MDL-06", "Commercial  MDL-00", "Commercial  MDL-06", "Commercial  MDL-94", "Commercial  MDL-96", "Exempt Condo", "Exmpt 490  MDL-00", "Exmpt Cm Cond OP", "Exmpt Cm Condo", "Exmpt Comm MDL-00", "Exmpt Comm  MDL-06", "Exmpt Comm  MDL-94", "Exmpt Comm  MDL-96", "Exmpt Ind", "Ind Condo  MDL-00", "Ind Condo  MDL-06", "Industrial  MDL-00", "Industrial  MDL-94", "Industrial  MDL-96")
```

In [9]:

```
#import geotags from GEOCODIO
geotags <- read.csv("Stamford addresses geotagged.csv")
```

In [10]:

```
#confirm Geotag loading
head (geotags)
```

Out[10]:

```
  Property.Address          Mailing.City Mailing.State Geocodio.Latitude
1 60 STRAWBERRY HILL AVENUE STAMFORD     CT            41.06237
2 60 STRAWBERRY HILL AVENUE STAMFORD     CT            41.06237
3 60 STRAWBERRY HILL AVENUE STAMFORD     CT            41.06237
4 107 LOCKWOOD AVENUE       STAMFORD     CT            41.05244
5 1400 BEDFORD STREET       STAMFORD     CT            41.06359
6 30 MAPLE TREE AVENUE      STAMFORD     CT            41.07158
  Geocodio.Longitude Geocodio.Accuracy.Score Geocodio.Accuracy.Type
1 -73.53354          1                       rooftop
2 -73.53354          1                       rooftop
3 -73.53354          1                       rooftop
4 -73.52190          1                       rooftop
5 -73.54079          1                       rooftop
6 -73.51507          1                       rooftop
  Geocodio.Address.Line.1 Geocodio.Address.Line.2 Geocodio.Address.Line.3 ⋯
1 60 Strawberry Hill Ave                          Stamford, CT 06902      ⋯
2 60 Strawberry Hill Ave                          Stamford, CT 06902      ⋯
3 60 Strawberry Hill Ave                          Stamford, CT 06902      ⋯
4 107 Lockwood Ave                                Stamford, CT 06902      ⋯
5 1400 Bedford St                                 Stamford, CT 06905      ⋯
6 30 Maple Tree Ave                               Stamford, CT 06906      ⋯
  Geocodio.Unit.Type Geocodio.Unit.Number Geocodio.City Geocodio.State
1                                         Stamford      CT
2                                         Stamford      CT
3                                         Stamford      CT
4                                         Stamford      CT
5                                         Stamford      CT
6                                         Stamford      CT
  Geocodio.County                    Geocodio.Postal.Code Geocodio.Country
1 Western Connecticut Planning Region 6902                 US
2 Western Connecticut Planning Region 6902                 US
3 Western Connecticut Planning Region 6902                 US
4 Western Connecticut Planning Region 6902                 US
5 Western Connecticut Planning Region 6905                 US
6 Western Connecticut Planning Region 6906                 US
  Geocodio.Source  Geocodio.Stable.Address.Key        Geocodio.Match.Type
1 City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9 building_centroid
2 City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9 building_centroid
3 City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9 building_centroid
4 City of Stamford gcod_usnjnsvcabu288cghmstrawx7pv8e building_centroid
```

```
5 City of Stamford gcod_usnxgapvae5gwy5dc6tzw8b72lcze building_centroid
6 City of Stamford gcod_usnlpm6xtnxdrsmear9lz6ghel7a9 building_centroid
```

In [11]:

```
#join geotags with grand list data
geotags_unique <- geotags %>%
    distinct(Property.Address, .keep_all = TRUE)
inner_joined <- inner_join(data, geotags_unique, by = "Property.Address")
```

In [12]:

```
#copy joined data for CPACE specific table
cpace_eligible_data <- inner_joined
```

In [13]:

```
#confirm CPACE load
head (cpace_eligible_data)
```

Out[13]:

```
  Parcel.ID Account.Number GIS.Tag    Map     Map.Cut Block Block.Cut Lot
1 16636     003-9031       W 181 8304     003 NA      9031  NA
2 16637     003-9032       W 181 8304     003 NA      9032  NA
3 16638     003-9033       W 181 8304     003 NA      9033  NA
4 16639     003-9034       W 181 8304     003 NA      9034  NA
5 16640     003-9035       W 181 8304     003 NA      9035  NA
6 16641     003-9036       W 181 8304     003 NA      9036  NA
  Lot.Cut Unit ⋯ Geocodio.Unit.Type Geocodio.Unit.Number Geocodio.City
1         618  ⋯                                         Stamford
2         702  ⋯                                         Stamford
3         703  ⋯                                         Stamford
4         704  ⋯                                         Stamford
5         705  ⋯                                         Stamford
6         706  ⋯                                         Stamford
  Geocodio.State Geocodio.County                     Geocodio.Postal.Code
1 CT             Western Connecticut Planning Region 6902
2 CT             Western Connecticut Planning Region 6902
3 CT             Western Connecticut Planning Region 6902
4 CT             Western Connecticut Planning Region 6902
5 CT             Western Connecticut Planning Region 6902
6 CT             Western Connecticut Planning Region 6902
  Geocodio.Country Geocodio.Source  Geocodio.Stable.Address.Key
1 US               City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9
2 US               City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9
3 US               City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9
4 US               City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9
5 US               City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9
6 US               City of Stamford gcod_usnte84mnkbs247ek43kzayzug2a9
  Geocodio.Match.Type
1 building_centroid
2 building_centroid
3 building_centroid
4 building_centroid
5 building_centroid
6 building_centroid
```

In [14]:

```
#confirm headers in cpace table
names(cpace_eligible_data)
```

Out[14]:

```
 [1] "Parcel.ID"
 [2] "Account.Number"
```

```
 [3] "GIS.Tag"
 [4] "Map"
 [5] "Map.Cut"
 [6] "Block"
 [7] "Block.Cut"
 [8] "Lot"
 [9] "Lot.Cut"
[10] "Unit"
[11] "Unit.Cut"
[12] "Internet.Suppression"
[13] "Town.ID"
[14] "Property.Address"
[15] "Property.City"
[16] "Property.State"
[17] "Property.Zip"
[18] "Property.County"
[19] "Street.Name"
[20] "Address.Number"
[21] "Address.Prefix"
[22] "Address.Suffix"
[23] "Owner"
[24] "Co.Owner"
[25] "Mailing.Address"
[26] "Mailing.Address.2"
[27] "Mailing.City.x"
[28] "Mailing.State.x"
[29] "Mailing.Zip"
[30] "Mailing.Address.Street.Name"
[31] "Mailing.Address.Street.Number"
[32] "Mailing.Address.Street.Prefix"
[33] "Mailing.Address.Street.Suffix"
[34] "Assessed.Total"
[35] "Assessed.Building"
[36] "Assessed.Land"
[37] "Assessed.Extra.Feature"
[38] "Assessed.Outbuilding"
[39] "Pre.Year.Assessed.Total"
[40] "Pre.Year.Assessed.Building"
[41] "Pre.Year.Assessed.Land"
[42] "Pre.Year.Assessed.Extra.Feature"
[43] "Pre.Year.Assessed.Outbuilding"
[44] "Appraised.Total"
[45] "Appraised.Building"
[46] "Appraised.Land"
[47] "Appraised.Extra.Feature"
[48] "Appraised.Outbuilding"
[49] "Valuation.Year"
[50] "Land.Acres"
[51] "Parcel.Depth"
[52] "Parcel.Frontage"
[53] "Water.Frontage.in.Feet"
[54] "Zone"
[55] "Zone.Description"
[56] "State.Use"
[57] "State.Use.Description"
[58] "Style.Description"
[59] "Model"
```

```
 [60] "Grade"
 [61] "Grade.Description"
 [62] "Depreciation"
 [63] "Condition"
 [64] "Condition.Description"
 [65] "Neighborhood"
 [66] "Actual.Year.Built"
 [67] "Effective.Year.Built"
 [68] "Replacement.Cost.New"
 [69] "Replacement.Cost.New.Less.Depreciation"
 [70] "Gross.Area.of.Primary.Building"
 [71] "Living.Area"
 [72] "Effective.Area"
 [73] "Stories"
 [74] "Frame.Type"
 [75] "Frame.Type.Description"
 [76] "Roof.Structure.Description"
 [77] "Roof.Cover.Description"
 [78] "Number.of.Buildings"
 [79] "Outbuilding.Count"
 [80] "Extra.Features.Count"
 [81] "Permit.Count"
 [82] "Total.Rooms"
 [83] "Number.of.Bedroom"
 [84] "Number.of.Bathrooms"
 [85] "Number.of.Half.Bathrooms"
 [86] "Bathroom.Style.Description"
 [87] "Kitchen.Style.Description"
 [88] "Family.Rooms"
 [89] "Attic.Type"
 [90] "Basement.Type"
 [91] "Interior.Floor.1.Description"
 [92] "Interior.Wall.1.Description"
 [93] "Exterior.Wall.1.Description"
 [94] "Heat.Type.Description"
 [95] "Heat.Fuel.Description"
 [96] "AC.Type"
 [97] "AC.Type.Description"
 [98] "Heat.AC"
 [99] "Heat.AC.Description"
[100] "Baths.Plumbing"
[101] "Baths.Plumbing.Description"
[102] "Ceiling.Wall"
[103] "Ceiling.Wall.Description"
[104] "Rooms.Partitions"
[105] "Rooms.Partitions.Description"
[106] "Percentage.of.Common.Wall"
[107] "Septic"
[108] "Well"
[109] "Number.of.Fireplaces"
[110] "Total.Extra.Fixtures"
[111] "Whirlpool"
[112] "Additional.Kitchen"
[113] "Basement.Garage"
[114] "Finished.Basement.Area"
[115] "Finished.Basement.Quality"
[116] "Base.Floor.Use..1st.floor."
```

```
[117] "Base.Floor.Use.Description..1st.floor."
[118] "Wall.Height"
[119] "Wall.Height.Description"
[120] "Book.Page"
[121] "Sale.Price"
[122] "Sale.Date"
[123] "Qualified"
[124] "Prior.Sale.Date"
[125] "Prior.Book.Page"
[126] "Prior.Sale.Price"
[127] "Sale.Grantee.Name"
[128] "Sale.Grantor.Name"
[129] "Instrument"
[130] "Validity.Code"
[131] "Occupancy"
[132] "Condo.Main.ID"
[133] "CAMA.Site.Link"
[134] "Building.Photo"
[135] "Number.of.Units"
[136] "Sketch"
[137] "Land.Use.Note"
[138] "Mailing.City.y"
[139] "Mailing.State.y"
[140] "Geocodio.Latitude"
[141] "Geocodio.Longitude"
[142] "Geocodio.Accuracy.Score"
[143] "Geocodio.Accuracy.Type"
[144] "Geocodio.Address.Line.1"
[145] "Geocodio.Address.Line.2"
[146] "Geocodio.Address.Line.3"
[147] "Geocodio.House.Number"
[148] "Geocodio.Street"
[149] "Geocodio.Unit.Type"
[150] "Geocodio.Unit.Number"
[151] "Geocodio.City"
[152] "Geocodio.State"
[153] "Geocodio.County"
[154] "Geocodio.Postal.Code"
[155] "Geocodio.Country"
[156] "Geocodio.Source"
[157] "Geocodio.Stable.Address.Key"
[158] "Geocodio.Match.Type"
```

In [15]:

```
#filter CPACE table
cpace_eligible_data <- cpace_eligible_data %>%
  filter(`State.Use.Description` %in% use_codes)
```

In [16]:

```
#confirm cpace table data
head (cpace_eligible_data)
```

Out[16]:

	Parcel.ID	Account.Number	GIS.Tag	Map	Map.Cut	Block	Block.Cut	Lot
1	17040	003-9732	W 181 8304	003	NA	9732	NA	
2	11632	003-1137	W 005 8304	003	NA	1137	NA	
3	11633	003-1138	W 005 8304	003	NA	1138	NA	
4	11638	003-1143	W 005 8304	003	NA	1143	NA	

```
5 11671     003-1291       S 070 2636     003 NA      1291  NA
6 28907     000-8870       S 003 4912     000 NA      8870  NA
  Lot.Cut Unit    ⋯ Geocodio.Unit.Type Geocodio.Unit.Number Geocodio.City
1             L-2 ⋯                                         Stamford
2         9       ⋯                                         Stamford
3         10      ⋯                                         Stamford
4         6       ⋯                                         Stamford
5                 ⋯                                         Stamford
6                 ⋯                                         Stamford
  Geocodio.State Geocodio.County                     Geocodio.Postal.Code
1 CT             Western Connecticut Planning Region 6902
2 CT             Western Connecticut Planning Region 6902
3 CT             Western Connecticut Planning Region 6902
4 CT             Western Connecticut Planning Region 6902
5 CT             Western Connecticut Planning Region 6902
6 CT             Western Connecticut Planning Region 6907
  Geocodio.Country Geocodio.Source
1 US               City of Stamford
2 US               City of Stamford
3 US               City of Stamford
4 US               City of Stamford
5 US               City of Stamford
6 US               TIGER/Line® dataset from the US Census Bureau
  Geocodio.Stable.Address.Key        Geocodio.Match.Type
1 gcod_usnte84mnkbs247ek43kzayzug2a9 building_centroid
2 gcod_usnte84mnkbs247etjlu4zfa74llf building_centroid
3 gcod_usnte84mnkbs247etjlu4zfa74llf building_centroid
4 gcod_usnte84mnkbs247etjlu4zfa74llf building_centroid
5 gcod_usntvhz3k7qw38eh9azcrnap7mrme building_centroid
6 gcod_usslsbpvdgesz4t2
```

In [ ]:

```
write.csv(filtered_data, "stamford_CPACE_potential.csv", row.names = FALSE)
```

In [17]:

```
#populate residential table
smarte_eligible_data <- inner_joined
```

In [18]:

```
#create residential filter
resi_use_codes <- list("Single Family")
```

In [19]:

```
#filter residential data
smarte_eligible_data <- smarte_eligible_data %>%
  filter(`State.Use.Description` %in% resi_use_codes)
```

In [ ]:

```
write.csv(smarte_eligible_data, "stamfords_smarte_potential.csv", row.names =
FALSE)
```

In [20]:

```
#Load tools for mapping
library(terra)
library(ggplot2)
library(sf)
library(ggspatial)
```

Out[20]:
```
terra 1.9.11

Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

In [21]:
```
#load adaptive capcity maps
map_data <- st_read("adaptiveCapacity.shp")
```

Out[21]:
```
Reading layer `adaptiveCapacity' from data source
  `/home/user/Capstone-Project-Template/adaptiveCapacity.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 44886 features and 28 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 1615904 ymin: 4623565 xmax: 1770904 ymax: 4697791
Projected CRS: NAD83 / UTM zone 16N
```

In [22]:
```
names (map_data)
```

Out[22]:
```
 [1] "OBJECTID"   "Join_Count" "TARGET_FID" "BUFF_DIST"  "ORIG_FID"
 [6] "cogs"       "city"       "Lon1"       "Lat1"       "x"
[11] "y"          "county"     "geomean"    "geomenSta"  "Coastal_st"
[16] "I95exits"   "Shelter_di" "disHospita" "geomean_1"  "geomenSta_"
[21] "ESI"        "distanceRi" "habitat"    "marsh_2055" "Shape_Leng"
[26] "Shape_Area" "geomeanAC"  "geomeanACN" "geometry"
```

In [31]:
```
#create CPACE mapping table
cpace_eligible_map <- cpace_eligible_data %>%
    select ('Property.Address','Geocodio.Longitude','Geocodio.Latitude')
```

In [32]:
```
#confirm new table
head (cpace_eligible_map)
```

Out[32]:
```
  Property.Address           Geocodio.Longitude Geocodio.Latitude
1 60 STRAWBERRY HILL AVENUE  -73.53354          41.06237
2 44 STRAWBERRY HILL AVENUE  -73.53409          41.06148
3 44 STRAWBERRY HILL AVENUE  -73.53409          41.06148
4 44 STRAWBERRY HILL AVENUE  -73.53409          41.06148
5 1051 EAST MAIN STREET      -73.51755          41.05658
6 0 LARKIN STREET            -73.52044          41.07814
```

In [33]:
```
#convert to shapefile
CPACE_map_SF <- st_as_sf(cpace_eligible_map, coords = c("Geocodio.Longitude",
"Geocodio.Latitude"), crs = 4326)
```

In [34]:
```
#transform from NAD83 to 4326
map_data_4326 <- st_transform(map_data, crs = 4326)
```

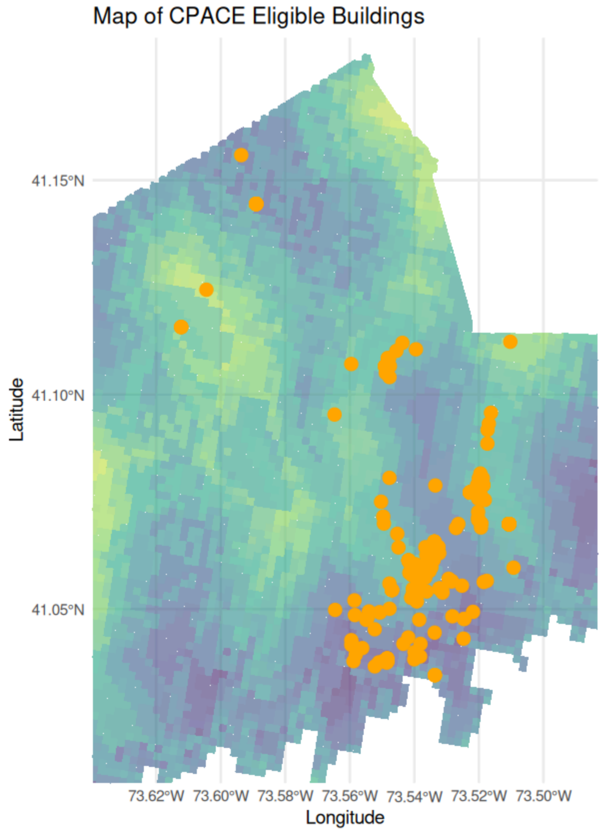
In [35]:
```
bbox_all <- st_bbox(CPACE_map_SF)
```

In [38]:
```
st_crs(CPACE_map_SF)
```

Out[38]:
```
Coordinate Reference System:
  User input: EPSG:4326
  wkt:
GEOGCRS["WGS 84",
    ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    USAGE[
        SCOPE["Horizontal component of 3D system."],
        AREA["World."],
        BBOX[-90,-180,90,180]],
    ID["EPSG",4326]]
```

In [48]:
```
# Plot the map with buildings
ggplot() +
    geom_sf(data = map_data_4326, aes(fill = geomeanACN), color = NA, alpha =
0.6) +  # Plot the map
    geom_sf(data = CPACE_map_SF , color = 'orange', size = 3) +  # Plot
buildings
    coord_sf(xlim = c(-73.632537, -73.490542),
           ylim = c(41.01755, 41.175326)) +
    #coord_sf(xlim = c(bbox_all["xmin"], bbox_all["xmax"]),
           #ylim = c(bbox_all["ymin"], bbox_all["ymax"])) +
  #annotation_map_tile(type = "osm") +  # adds OpenStreetMap tiles
    #coord_sf(xlim = c(-73.632537, -73.490542),
           #ylim = c(41.01755, 41.175326)) +
  scale_fill_viridis_c() +
  theme_minimal() +
  labs(title = "Map of CPACE Eligible Buildings",
       x = "Longitude",
       y = "Latitude") +
  theme(legend.position = "none")
```

Out[48]:


Map of CPACE Eligible Buildings
41.15°N
41.10°N
41.05°N
Latitude
73.62°W 73.60°W 73.58°W 73.56°W 73.54°W 73.52°W 73.50°W
Longitude